

Auteur: Sabawoon Enayat
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Oefeningenreeks 4A nr. 42.2

$$\begin{aligned}\text{Opgave: } I &= \int \frac{dx}{x^2 - 4} \\ &= \int \frac{dx}{(x-2)(x+2)} \\ \Rightarrow \frac{1}{(x-2)(x+2)} &= \frac{A}{x-2} + \frac{B}{x+2} \\ A &= \left. \frac{1}{x+2} \right|_{x=2} = \frac{1}{4} \\ B &= \left. \frac{1}{x-2} \right|_{x=-2} = -\frac{1}{4} \\ I &= \frac{1}{4} \left(\int \frac{dx}{x-2} - \int \frac{dx}{x+2} \right) \\ &= \frac{1}{4} \left(\int \frac{d(x-2)}{x-2} - \int \frac{d(x+2)}{x+2} \right) \\ &= \frac{1}{4} (\ln|x-2| - \ln|x+2|) + c \\ &= \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| + c\end{aligned}$$

References